



DEPARTMENT OF ENGINEERING
INFORMATION AND COMPUTER

SIGNALS AND SYSTEMS

TASK 3
CONSTANT TIME SYSTEMS

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Student Semester: 4th

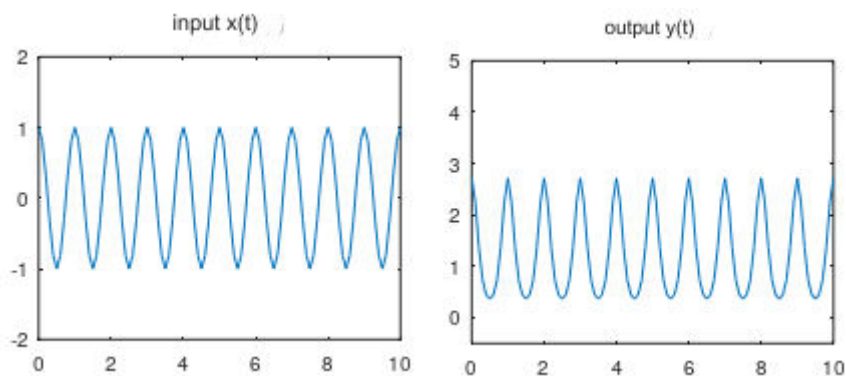
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EXERCISE 2

Commands:

```
t = 0:0.1:10;  
x = cos(2*pi*t);  
subplot( 2,2,1 );  
plot( t,x )  
title( ' input x(t) ' );  
ylim ( [-2,2])  
y = exp(x);  
subplot( 2,2,2 );  
plot( t,y )  
title( ' output y(t) ' );  
ylim([-0.5,5]);
```

Graphic representations:



Documentation:

First, we will take as an input signal a signal $x(t) = \cos(2\pi t)$.

We observe that for $-1 \leq x(t) \leq 1$, the input signal $x(t)$ is blocked for $M = 1$ since $|x(t)| \leq M$. Also, we observe that for $-0.5 \leq y(t) \leq 2.5$, the output signal $y(t)$ is blocked for $N = 1.5$, since $|y(t)| \leq N$.

Therefore the system with input-output relation $y(t) = e^{x(t)}$ is

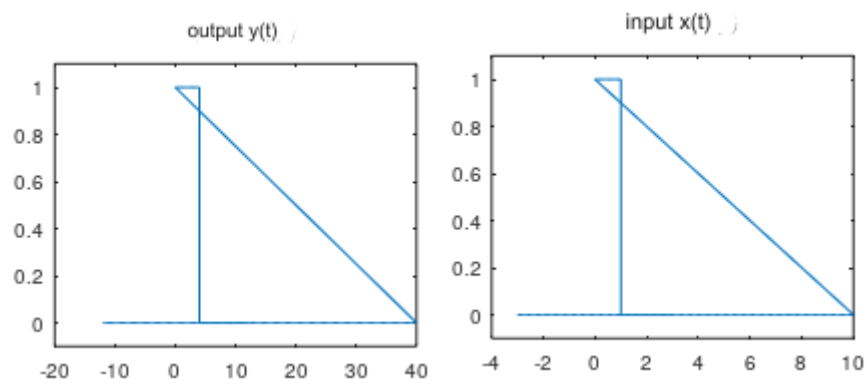
stable.

EXERCISE 3

Commands :

```
t1 = -3:0.1:10;  
x1 = zeros(size(t1));  
t2 = 0:0.1:1;  
x2 = ones(size(t2));  
t3 = 1:0.1:3;  
x3 = zeros(size(t3));  
t = [t1 t2 t3];  
x = [x1 x2 x3];  
subplot( 2,2,1 );  
plot( t,x )  
title( ' input x(t)' );  
ylim ( [-0.1,1.1]);  
subplot( 2,2,2 );  
plot( t*4,x );  
title( ' output y(t)' );  
ylim ( [-0.1,1.1]);
```

Graphic representations:



Documentation:

In principle, we will take as an input signal a signal $x(t) = u(t) - u(t - 1)$.

We notice that the output $y(t) = x(t/4)$ appears in a time shorter than that of the input. Therefore, the system is non-causal.